

2011. Final Value of Variable After Performing Operations

There is a programming language with only **four** operations and **one** variable `x`:

- `++x` and `x++` **increments** the value of the variable `x` by `1`.
- `--x` and `x--` **decrements** the value of the variable `x` by `1`.

Initially, the value of `x` is `0`.

Given an array of strings `operations` containing a list of operations, return *the **final** value of `x` after performing all the operations.*

Example 1:

```
Input: operations = ["--X","X++","X++"]
Output: 1
Explanation: The operations are performed as follows:
Initially, X = 0.
--X: X is decremented by 1, X = 0 - 1 = -1.
X++: X is incremented by 1, X = -1 + 1 = 0.
X++: X is incremented by 1, X = 0 + 1 = 1.
```

Example 2:

```
Input: operations = ["++X","++X","X++"]
Output: 3
Explanation: The operations are performed as follows:
Initially, X = 0.
++X: X is incremented by 1, X = 0 + 1 = 1.
++X: X is incremented by 1, X = 1 + 1 = 2.
X++: X is incremented by 1, X = 2 + 1 = 3.
```

Example 3:

```
Input: operations = ["X++","++X","--X","X--"]
Output: 0
Explanation: The operations are performed as follows:
```

```
Initially, X = 0.  
X++: X is incremented by 1, X = 0 + 1 = 1.  
++X: X is incremented by 1, X = 1 + 1 = 2.  
--X: X is decremented by 1, X = 2 - 1 = 1.  
X--: X is decremented by 1, X = 1 - 1 = 0.
```

Constraints:

- `1 <= operations.length <= 100`

Solution: time = $O(n)$ and space = $O(1)$

```
class Solution:  
    def finalValueAfterOperations(self, operations: List[str]) -> int:  
        res = 0  
        for i in operations:  
            if i == '++X' or i == 'X++':  
                res += 1  
            else:  
                res -= 1  
        return res
```

same but pythonic

```
class Solution:  
    def finalValueAfterOperations(self, operations: List[str]) -> int:  
        res = 0  
        for op in operations:  
            res += 1 if '+' in op else -1  
        return res
```

in oneline

```
class Solution:  
    def finalValueAfterOperations(self, operations: List[str]) -> int:  
        return sum(1 if '+' in op else -1 for op in operations)
```

```
r]) -> int:  
    return sum(1 if '+' in op else -1 for op in operation  
s)
```